CALIBRATION SHEETS

SBE 37IM Configuration - S/N 7322..... 1

SBE 37IM Pressure Test Certificate - S/N 7322..... 2

SBE 37IM Temperature Calibration - S/N 7322..... 3

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SBE 37IM Pressure Calibration - S/N 7322..... 5

SBE 37IM Configuration - S/N 7323..... 6

SBE 37IM Pressure Test Certificate - S/N 7323..... 7

SBE 37IM Temperature Calibration - S/N 7323..... 8

SBE 37IM Conductivity Calibration - S/N 7323..... 9

SBE 37IM Pressure Calibration - S/N 7323..... 10

SBE 37IM Configuration - S/N 7324..... 11

SBE 37IM Pressure Test Certificate - S/N 7324..... 12

SBE 37IM Temperature Calibration - S/N 7324..... 13

SBE 37IM Conductivity Calibration - S/N 7324..... 14

SBE 37IM Pressure Calibration - S/N 7324..... 15

SBE 37IM Configuration - S/N 7325..... 16

CALIBRATION SHEETS

SBE 37IM Pressure Test Certificate - S/N 7325..... 17

SBE 37IM Temperature Calibration - S/N 7325..... 18

SBE 37IM Conductivity Calibration - S/N 7325..... 19

SBE 37IM Pressure Calibration - S/N 7325..... 20

SBE 37IM Configuration - S/N 7326..... 21

SBE 37IM Pressure Test Certificate - S/N 7326..... 22

SBE 37IM Temperature Calibration - S/N 7326..... 23

SBE 37IM Conductivity Calibration - S/N 7326..... 24

SBE 37IM Pressure Calibration - S/N 7326..... 25

SBE 37IM Configuration - S/N 7327..... 26

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SBE 37IM Temperature Calibration - S/N 7327..... 28

SBE 37IM Conductivity Calibration - S/N 7327..... 29

SBE 37IM Pressure Calibration - S/N 7327..... 30

SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM53517-7322
Pressure Sensor	2000 Dbar Druck, SN 2956219
Firmware Version	V 3.0b / V 2.0a
Memory	8192Kb
Conductivity Range	0-7 S/m
Microcat ID#	01
Zero Conductivity	2559.242 Hz
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**2000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
Website: http://www.seabird.com

Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **8/21/2009** Description **SBE-37 Microcat**

Job Number: **53517** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **7322**

Pressure Sensor Information:

Sensor Type: **Druck**
Sensor Serial Number: **2956219**
Sensor Rating: **2900**

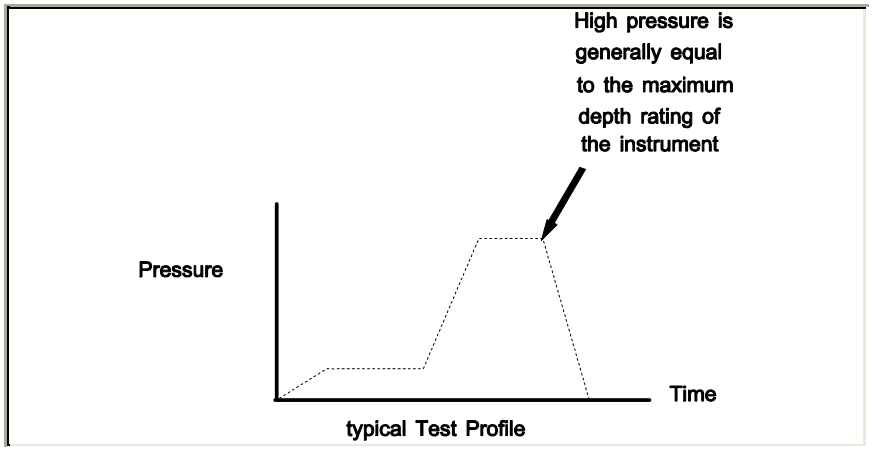
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **2900** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: VG



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7322
CALIBRATION DATE: 15-Sep-09

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

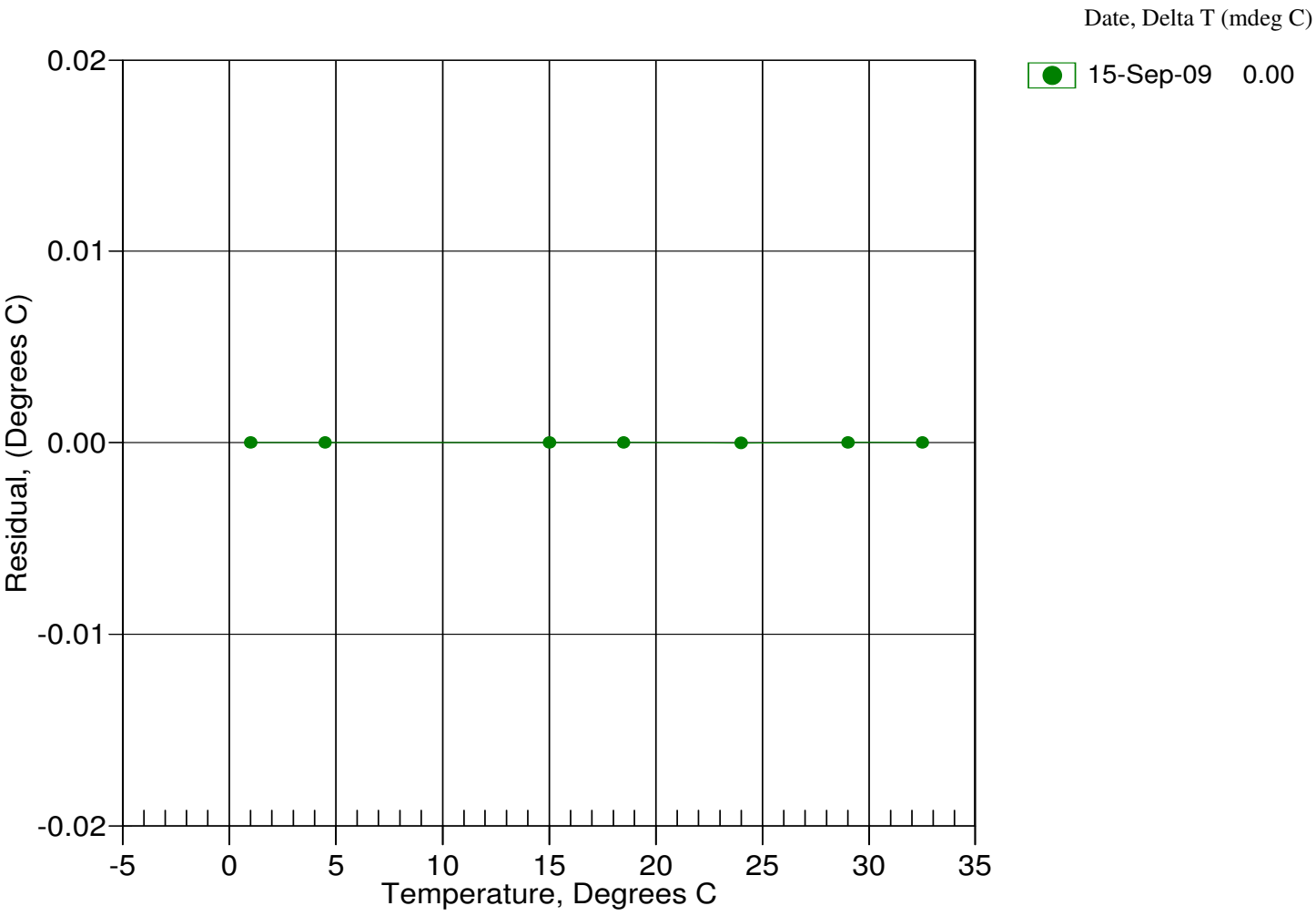
ITS-90 COEFFICIENTS

a0 = -7.856420e-005
a1 = 2.933755e-004
a2 = -3.117387e-006
a3 = 1.634230e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	579898.0	1.0000	0.0000
4.5000	496582.0	4.5000	-0.0000
15.0000	318014.0	15.0000	-0.0000
18.5000	275829.0	18.5000	0.0000
24.0000	221875.0	24.0000	-0.0000
29.0000	183153.0	29.0000	0.0000
32.5000	160676.0	32.5000	-0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



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Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7322
CALIBRATION DATE: 15-Sep-09

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.017731e+000
h =  1.556709e-001
i = -2.039632e-004
j =  3.900327e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  0.0000e+000
```

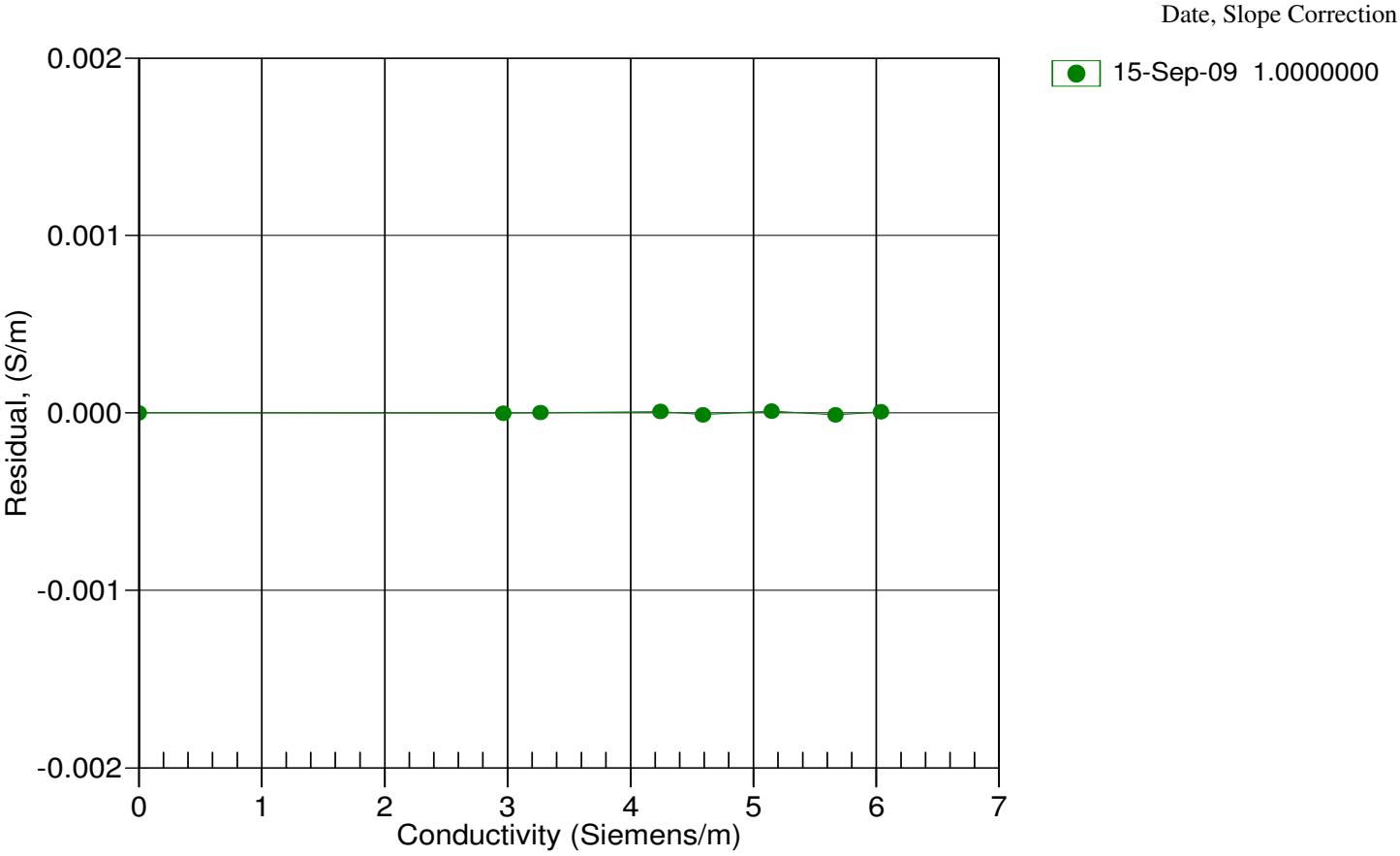
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2559.09	0.00000	0.00000
1.0000	34.6526	2.96330	5057.57	2.96330	-0.00000
4.5000	34.6328	3.26911	5247.62	3.26911	0.00000
15.0000	34.5895	4.24672	5812.96	4.24673	0.00001
18.5000	34.5800	4.59039	5998.80	4.59038	-0.00001
24.0000	34.5691	5.14590	6287.35	5.14591	0.00001
29.0000	34.5631	5.66550	6545.43	5.66548	-0.00001
32.5000	34.5598	6.03630	6723.40	6.03631	0.00001

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7322
CALIBRATION DATE: 10-Sep-09

SBE 37 PRESSURE CALIBRATION DATA
2900 psia S/N 2956219

COEFFICIENTS:

PA0 =	4.290880e-001	PTCA0 =	5.246135e+005
PA1 =	9.328526e-003	PTCA1 =	3.737034e+000
PA2 =	-1.767194e-010	PTCA2 =	-7.123853e-002
PTEMPA0 =	-7.363470e+001	PTCB0 =	2.522075e+001
PTEMPA1 =	4.829592e-002	PTCB1 =	1.500000e-004
PTEMPA2 =	-1.897768e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FSR
14.73	526194.0	1961.0	14.74	0.00
614.95	590625.0	1965.0	614.94	-0.00
1215.01	655194.0	1966.0	1214.96	-0.00
1815.09	719927.0	1967.0	1815.02	-0.00
2415.11	784816.0	1967.0	2415.04	-0.00
3015.10	849869.0	1969.0	3015.09	-0.00
2415.05	784829.0	1967.0	2415.16	0.00
1815.04	719930.0	1967.0	1815.05	0.00
1214.92	655199.0	1966.0	1215.00	0.00
614.96	590626.0	1966.0	614.95	-0.00
14.74	526194.0	1968.0	14.73	-0.00

THERMAL CORRECTION

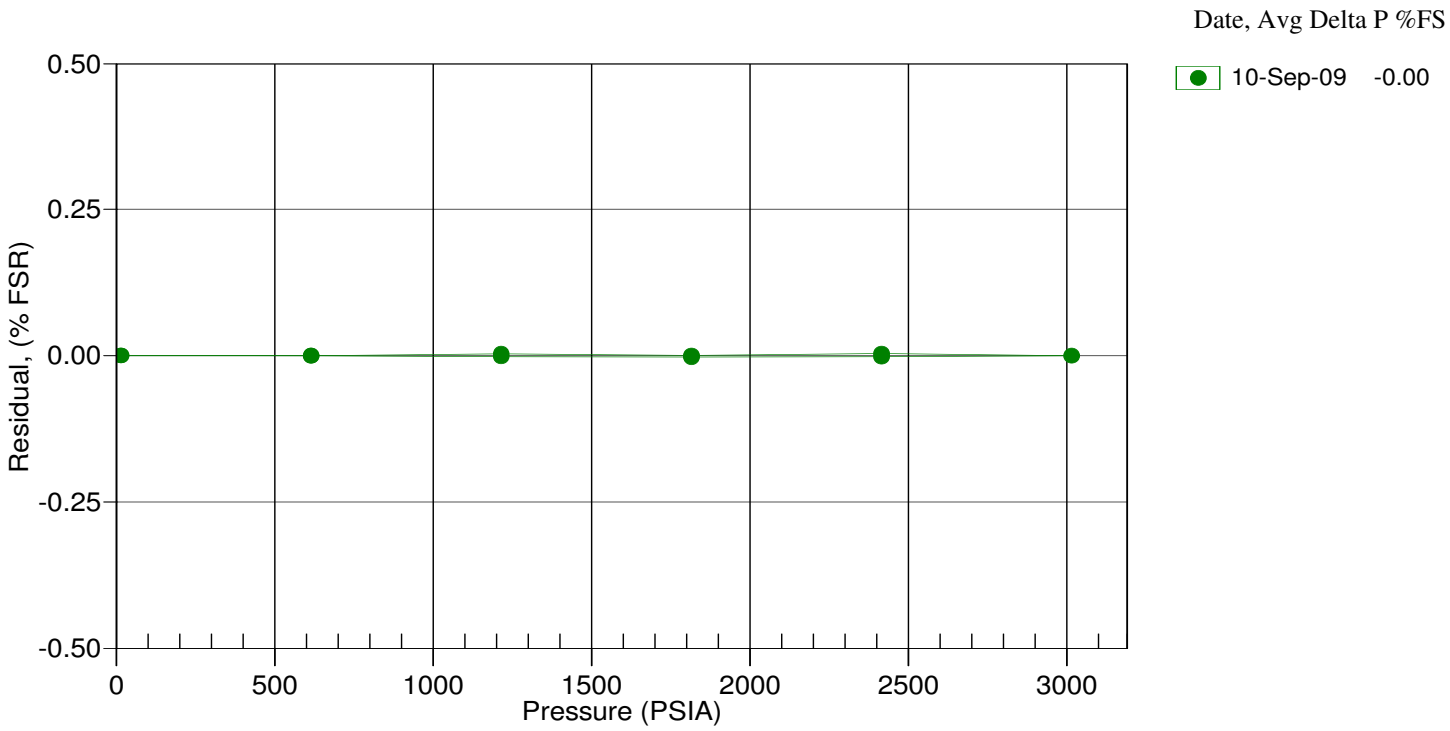
TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2217	526276.00
29.00	2143	526280.95
24.00	2038	526282.35
18.50	1922	526275.27
15.00	1849	526269.45
4.50	1628	526247.70
1.00	1555	526234.79

TEMP(ITS90)	SPAN(mV)
-5.00	25.22
35.00	25.23

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM53517-7323
Pressure Sensor	2000 Dbar Druck, SN 2956228
Firmware Version	V 3.0b / V 2.0a
Memory	8192Kb
Conductivity Range	0-7 S/m
Microcat ID#	02
Zero Conductivity	2608.578 Hz
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**2000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
Website: http://www.seabird.com

Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **8/21/2009** Description **SBE-37 Microcat**

Job Number: **53517** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **7323**

Pressure Sensor Information:

Sensor Type: **Druck**
Sensor Serial Number: **2956228**
Sensor Rating: **2900**

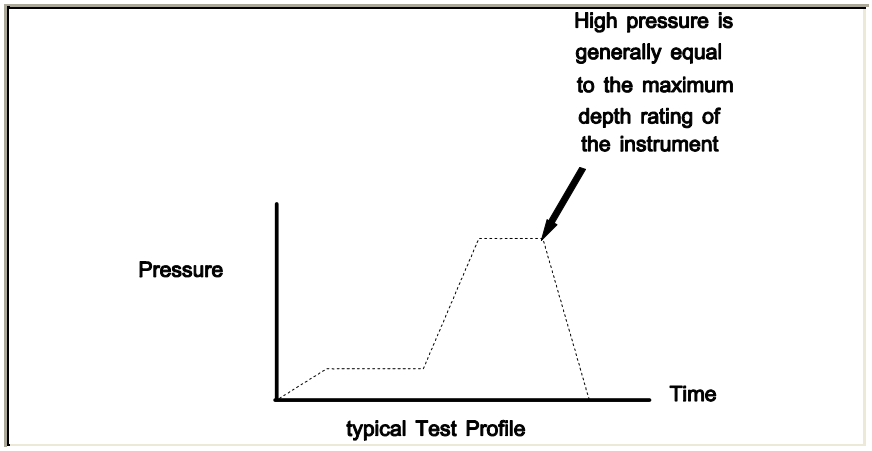
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **2900** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: VG



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7323
CALIBRATION DATE: 15-Sep-09

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

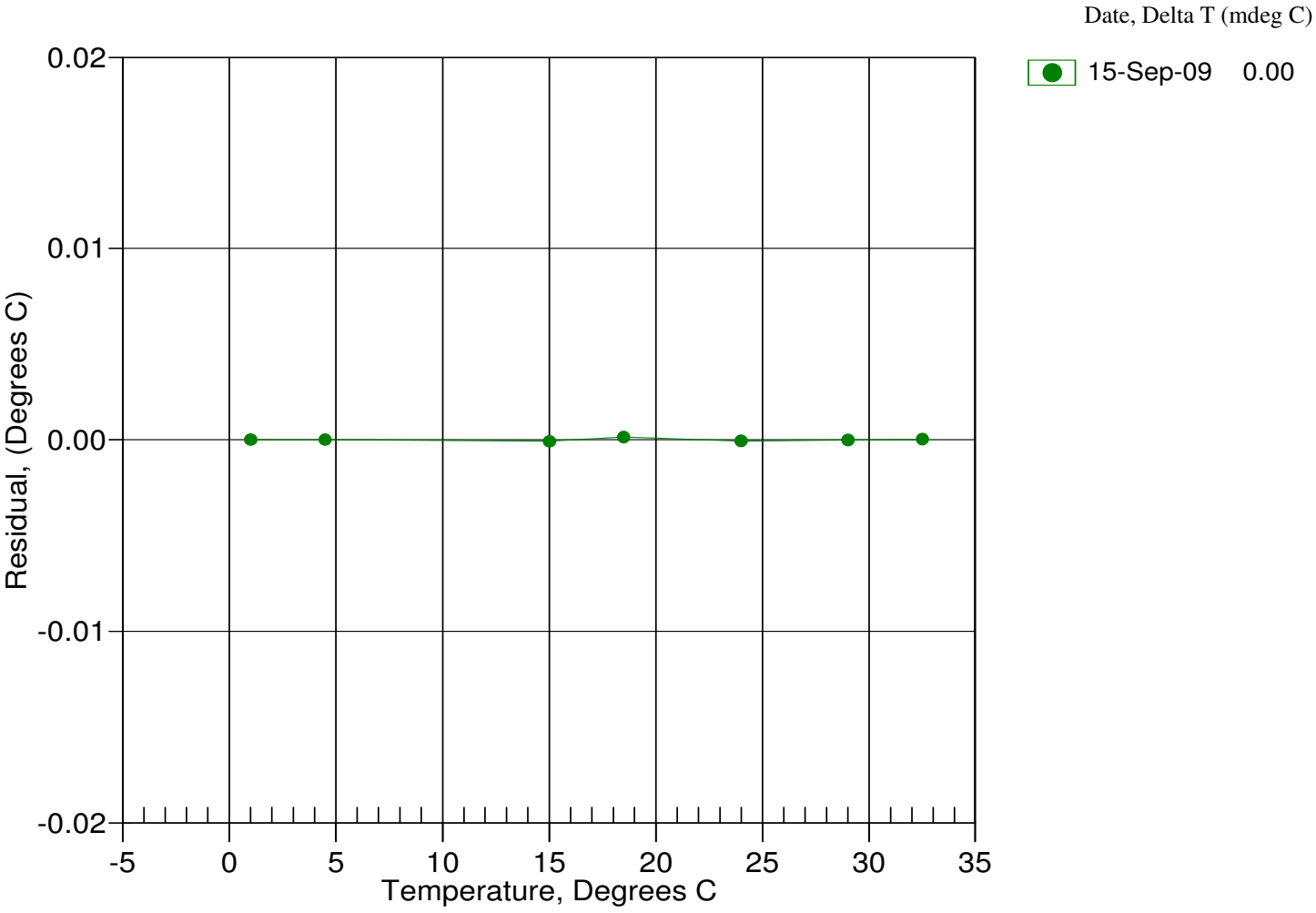
ITS-90 COEFFICIENTS

a0 = -6.804369e-005
a1 = 2.913288e-004
a2 = -3.011641e-006
a3 = 1.631421e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	577319.0	1.0000	0.0000
4.5000	494520.0	4.5000	-0.0000
15.0000	316941.0	14.9999	-0.0001
18.5000	274958.0	18.5001	0.0001
24.0000	221247.0	23.9999	-0.0001
29.0000	182682.0	29.0000	-0.0000
32.5000	160289.0	32.5000	0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



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Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7323
CALIBRATION DATE: 15-Sep-09

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.030214e+000
h =  1.518906e-001
i = -2.983649e-004
j =  4.417143e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  0.0000e+000
```

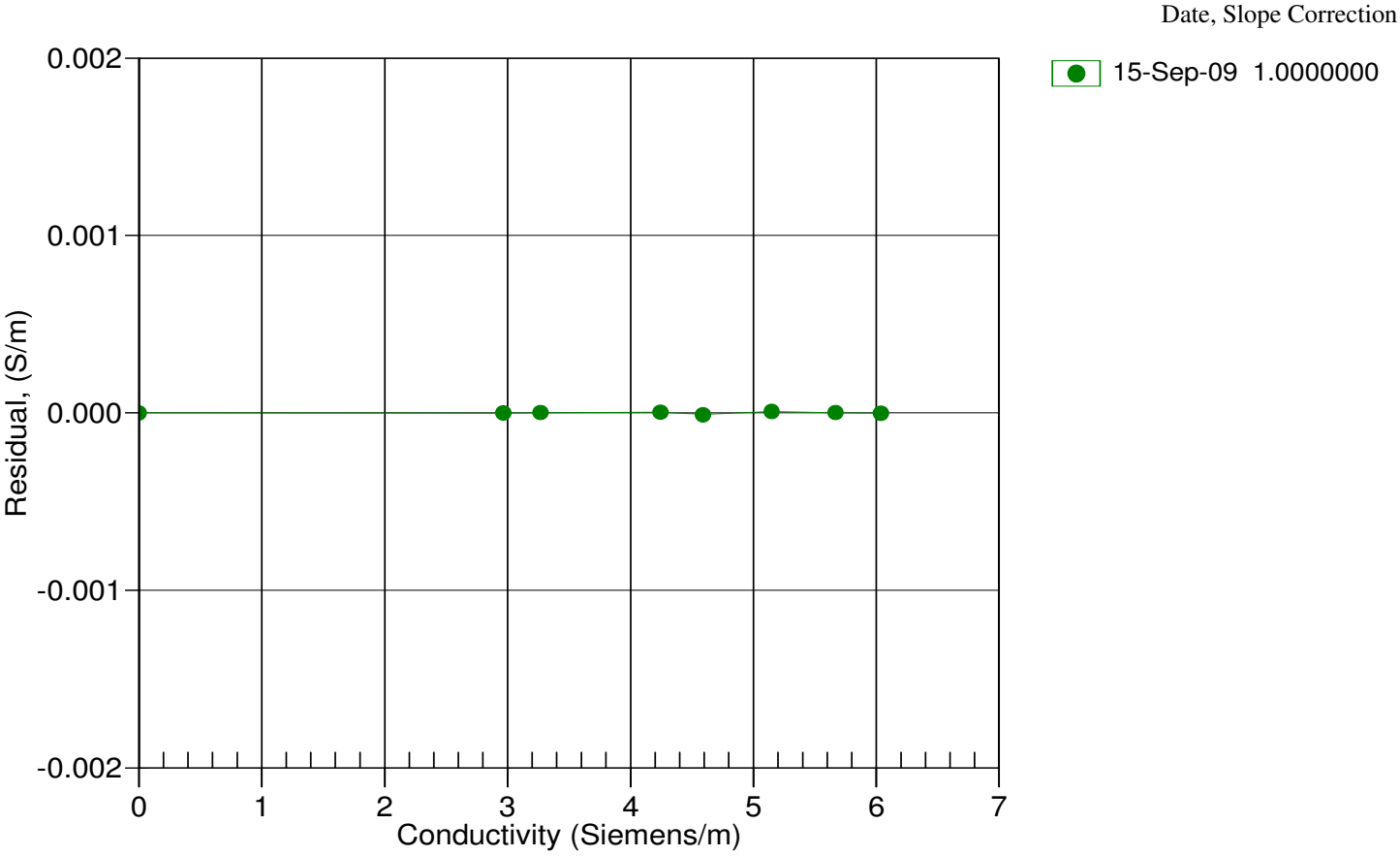
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2608.45	0.00000	0.00000
1.0000	34.6526	2.96330	5133.80	2.96330	-0.00000
4.5000	34.6328	3.26911	5326.21	3.26911	0.00000
15.0000	34.5895	4.24672	5898.66	4.24673	0.00000
18.5000	34.5800	4.59039	6086.86	4.59038	-0.00001
24.0000	34.5691	5.14590	6379.08	5.14591	0.00001
29.0000	34.5631	5.66550	6640.46	5.66550	0.00000
32.5000	34.5598	6.03630	6820.69	6.03630	-0.00000

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7323
CALIBRATION DATE: 10-Sep-09

SBE 37 PRESSURE CALIBRATION DATA
2900 psia S/N 2956228

COEFFICIENTS:

PA0 =	2.597497e-001	PTCA0 =	5.247969e+005
PA1 =	9.471979e-003	PTCA1 =	2.413177e+000
PA2 =	-1.876493e-010	PTCA2 =	-5.339543e-002
PTEMPA0 =	-7.211111e+001	PTCB0 =	2.523412e+001
PTEMPA1 =	4.987193e-002	PTCB1 =	-1.750000e-004
PTEMPA2 =	-5.550526e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FSR
14.73	526353.0	1899.0	14.74	0.00
614.95	589794.0	1902.0	614.95	0.00
1215.01	653376.0	1903.0	1214.97	-0.00
1815.09	717125.0	1904.0	1815.05	-0.00
2415.11	781032.0	1904.0	2415.09	-0.00
3015.10	845099.0	1905.0	3015.08	-0.00
2415.05	781035.0	1904.0	2415.11	0.00
1815.04	717125.0	1904.0	1815.05	0.00
1214.92	653374.0	1903.0	1214.96	0.00
614.96	589792.0	1903.0	614.93	-0.00
14.74	526352.0	1903.0	14.73	-0.00

THERMAL CORRECTION

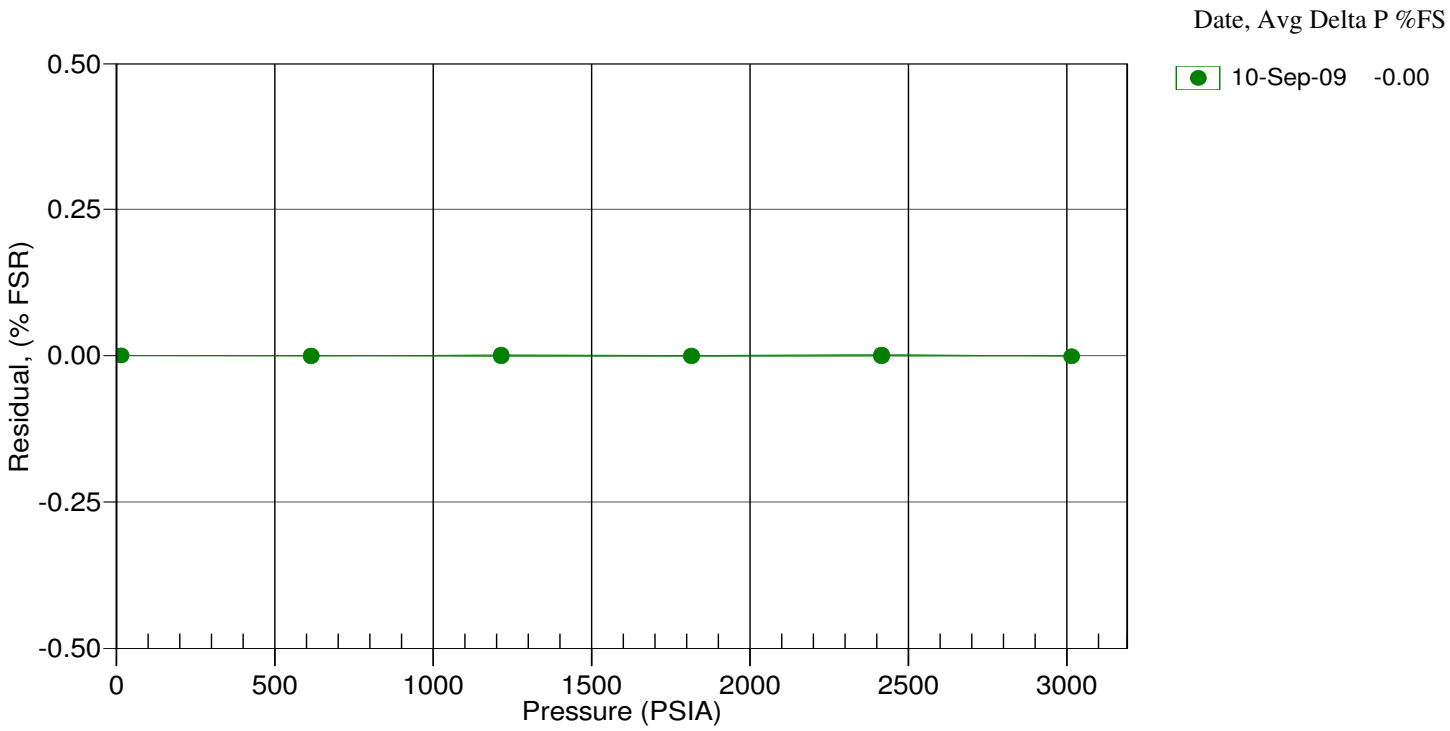
TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2149	526431.00
29.00	2075	526437.97
24.00	1971	526439.41
18.50	1855	526437.35
15.00	1782	526432.55
4.50	1563	526421.82
1.00	1491	526412.92

TEMP(ITS90)	SPAN(mV)
-5.00	25.23
35.00	25.23

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM53517-7324
Pressure Sensor	1000 Dbar Druck, SN 2895333
Firmware Version	V 3.0b / V 2.0a
Memory	8192Kb
Conductivity Range	0-7 S/m
Microcat ID#	03
Zero Conductivity	2580.441 Hz
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**1000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
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Website: http://www.seabird.com

Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **8/24/2009** Description **SBE-37 Microcat**

Job Number: **53517** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **7324**

Pressure Sensor Information:

Sensor Type: **Druck**
Sensor Serial Number: **2895333**
Sensor Rating: **1450**

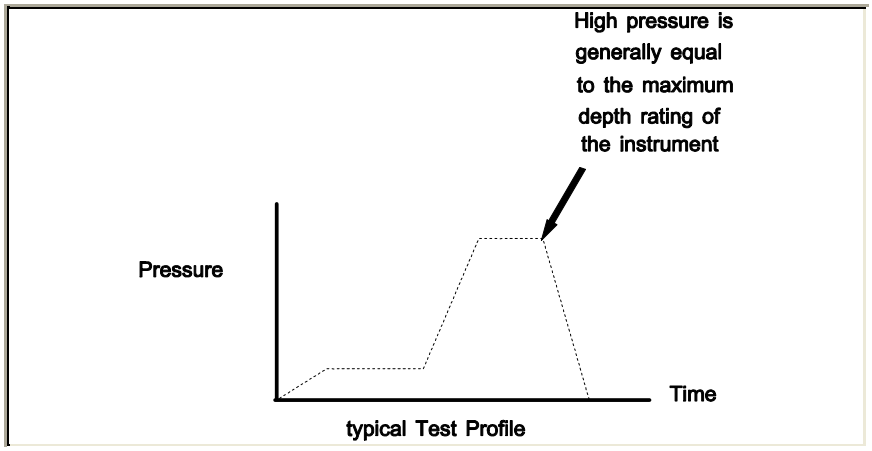
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **1450** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: ND



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7324
CALIBRATION DATE: 15-Sep-09

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

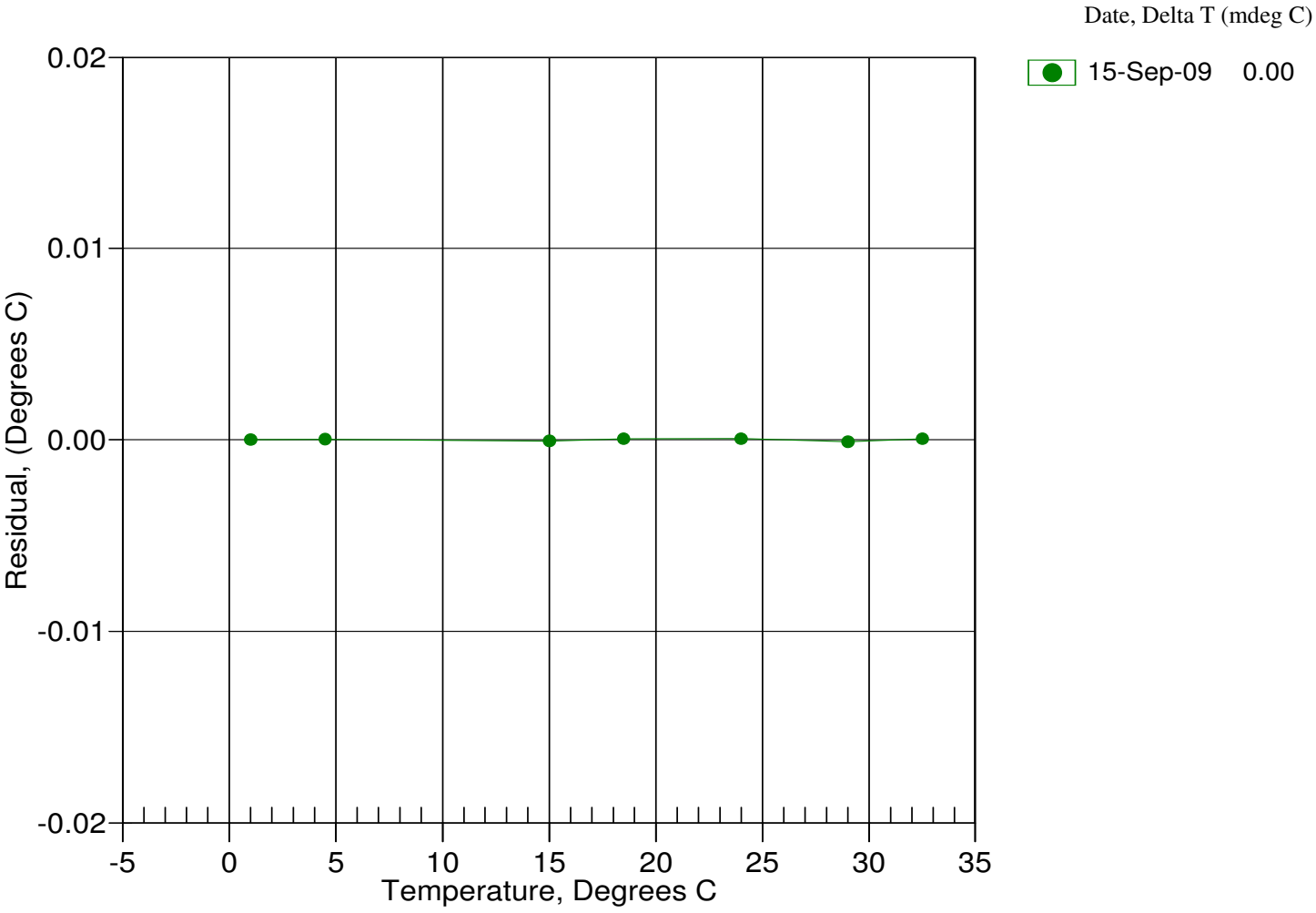
ITS-90 COEFFICIENTS

a0 = -9.695827e-005
a1 = 2.981402e-004
a2 = -3.492542e-006
a3 = 1.754330e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	566747.0	1.0000	-0.0000
4.5000	485568.0	4.5000	0.0000
15.0000	311402.0	14.9999	-0.0001
18.5000	270211.0	18.5000	0.0000
24.0000	217498.0	24.0001	0.0001
29.0000	179643.0	28.9999	-0.0001
32.5000	157656.0	32.5000	0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7324
CALIBRATION DATE: 15-Sep-09

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.011890e+000
h =  1.522861e-001
i = -2.166856e-004
j =  3.847063e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  2.5835e-007
```

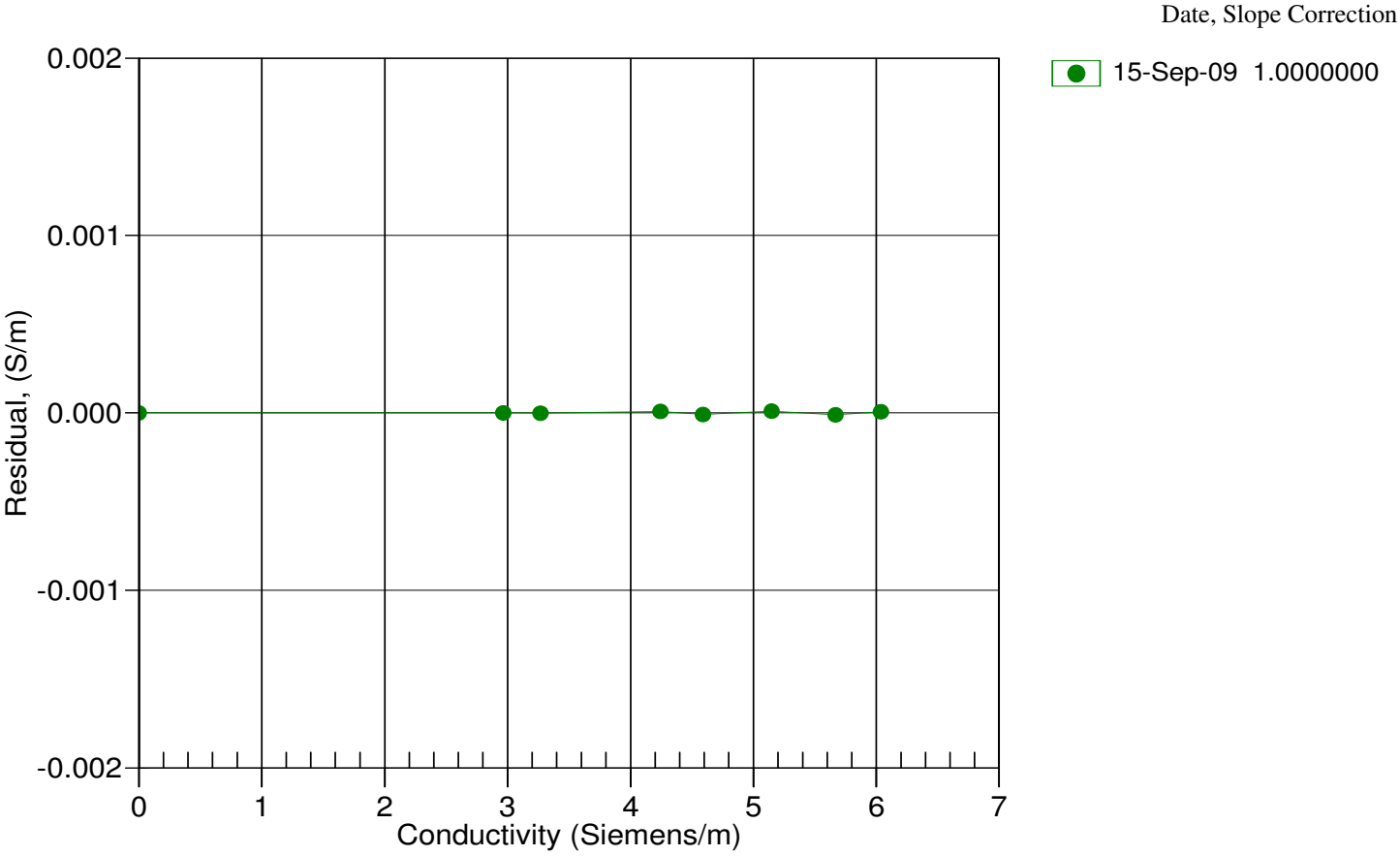
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2580.28	0.00000	0.00000
1.0000	34.6526	2.96330	5110.88	2.96330	0.00000
4.5000	34.6328	3.26911	5303.23	3.26911	-0.00000
15.0000	34.5895	4.24672	5875.37	4.24673	0.00001
18.5000	34.5800	4.59039	6063.43	4.59038	-0.00001
24.0000	34.5691	5.14590	6355.41	5.14591	0.00001
29.0000	34.5631	5.66550	6616.55	5.66549	-0.00001
32.5000	34.5598	6.03630	6796.62	6.03630	0.00000

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7324
CALIBRATION DATE: 09-Sep-09

SBE 37 PRESSURE CALIBRATION DATA
1450 psia S/N 2895333

COEFFICIENTS:

PA0 =	1.346481e-001	PTCA0 =	5.249521e+005
PA1 =	4.601907e-003	PTCA1 =	3.744328e+000
PA2 =	-1.534422e-011	PTCA2 =	-1.283200e-001
PTEMPA0 =	-7.165370e+001	PTCB0 =	2.529337e+001
PTEMPA1 =	4.934437e-002	PTCB1 =	-5.250000e-004
PTEMPA2 =	-3.859024e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FSR
14.62	528121.0	1913.0	14.62	0.00
314.97	593353.0	1931.0	314.89	-0.01
614.88	658547.0	1933.0	614.85	-0.00
914.86	723782.0	1933.0	914.86	-0.00
1214.90	789034.0	1934.0	1214.82	-0.01
1464.86	843449.0	1935.0	1464.86	-0.00
1214.80	789042.0	1934.0	1214.85	0.00
914.78	723779.0	1936.0	914.85	0.00
614.82	658544.0	1936.0	614.84	0.00
314.87	593355.0	1937.0	314.90	0.00
14.63	528125.0	1941.0	14.66	0.00

THERMAL CORRECTION

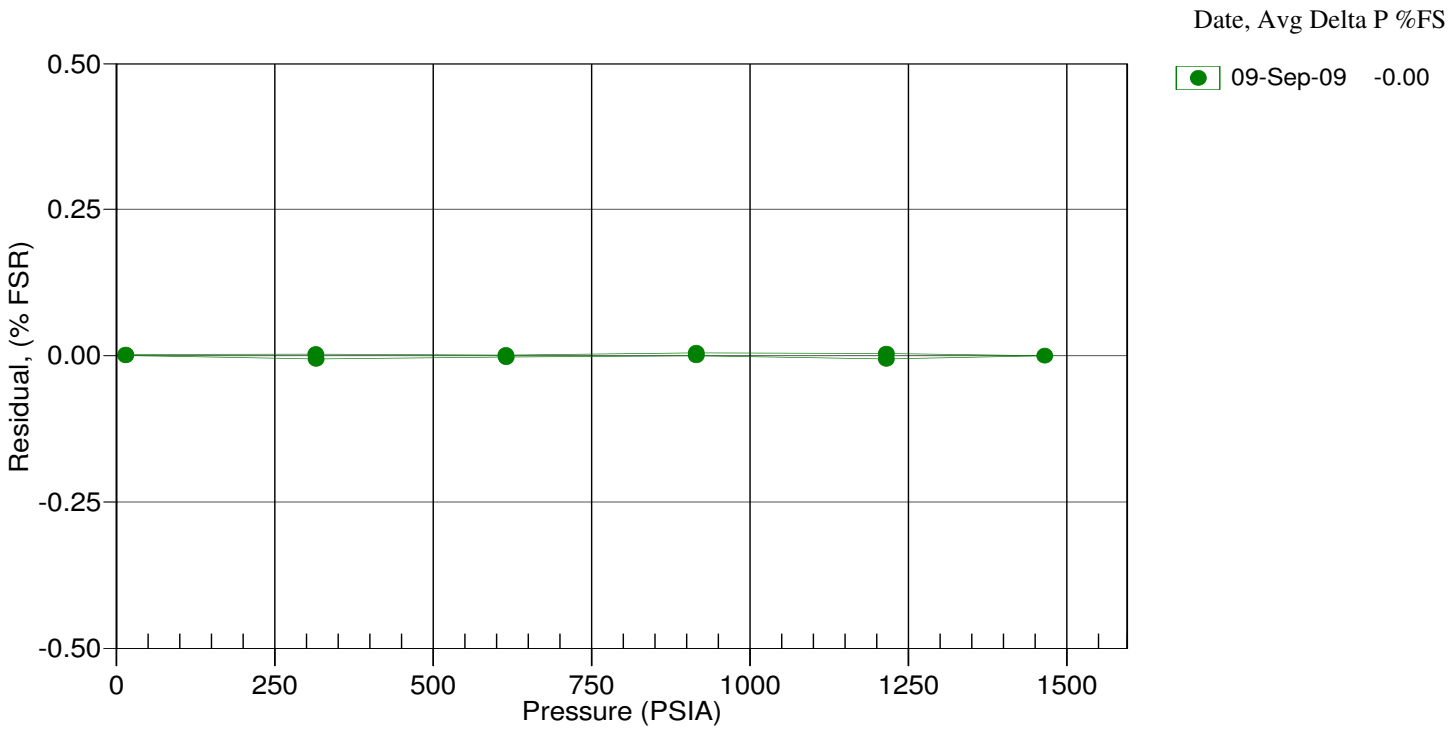
TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2147	528286.00
29.00	2073	528301.94
24.00	1969	528315.64
18.50	1854	528324.36
15.00	1781	528327.84
4.50	1562	528316.34
1.00	1490	528302.43

TEMP(ITS90)	SPAN(mV)
-5.00	25.30
35.00	25.27

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM53517-7325
Pressure Sensor	1000 Dbar Druck, SN 2895334
Firmware Version	V 3.0b / V 2.0a
Memory	8192Kb
Conductivity Range	0-7 S/m
Microcat ID#	04
Zero Conductivity	2863.469 Hz
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**1000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
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Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **8/24/2009** Description **SBE-37 Microcat**

Job Number: **53517** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **7325**

Pressure Sensor Information:

Sensor Type: **Druck**
Sensor Serial Number: **2895334**
Sensor Rating: **1450**

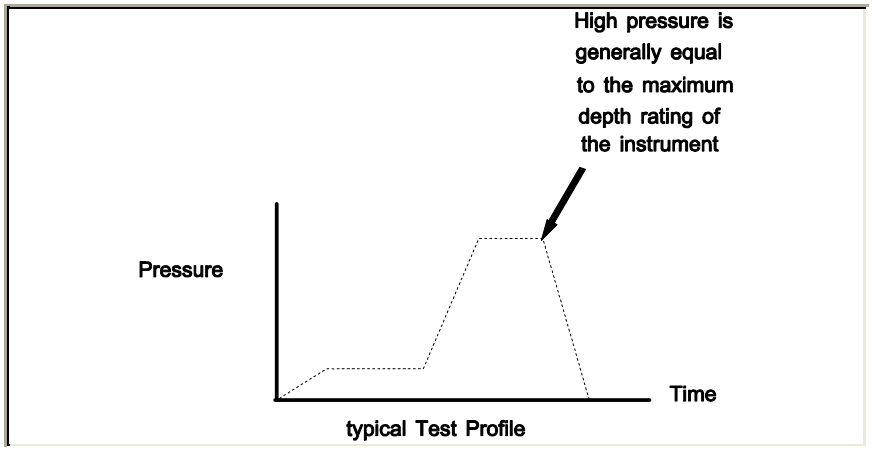
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **1450** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: ND



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7325
CALIBRATION DATE: 15-Sep-09

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

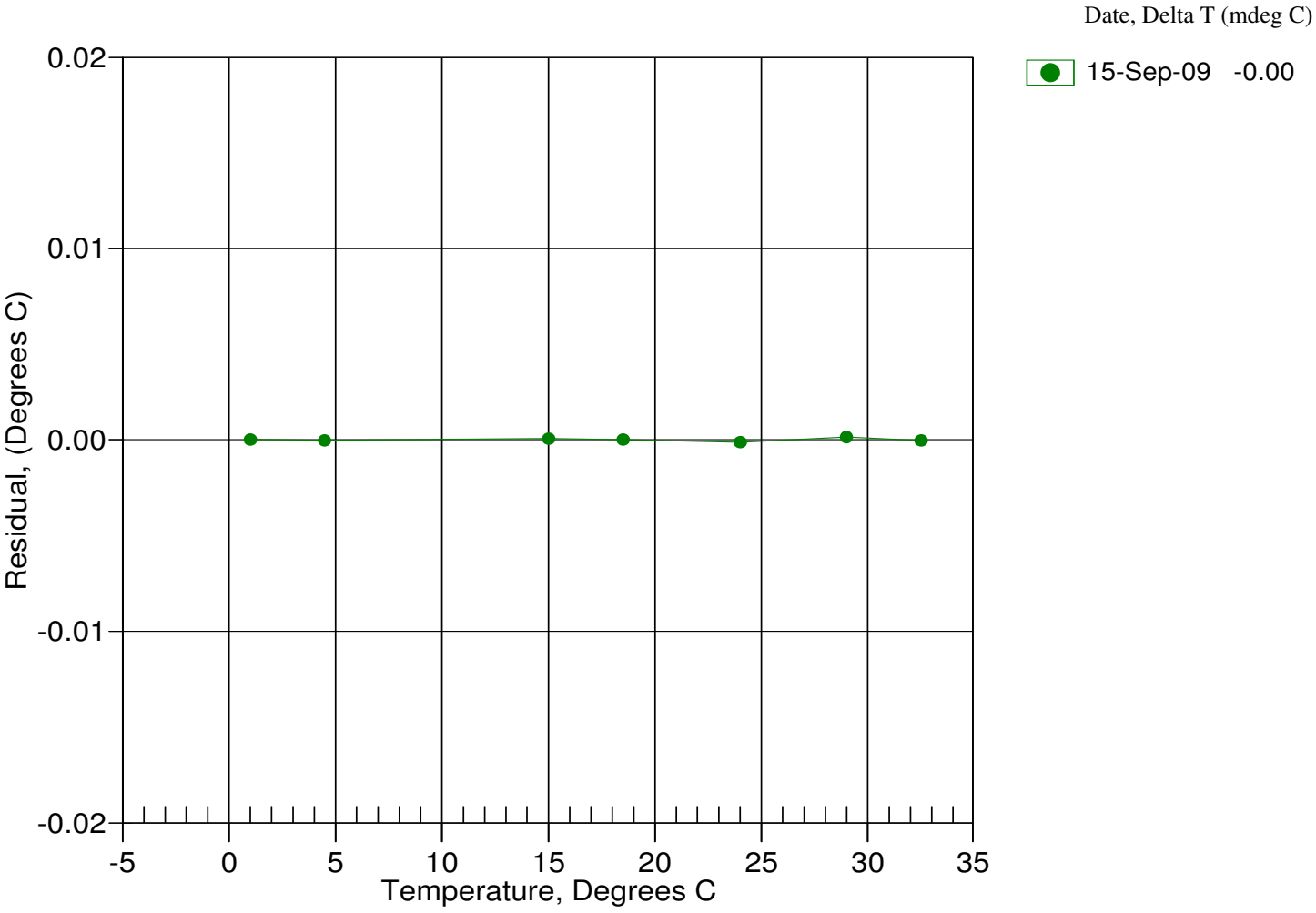
ITS-90 COEFFICIENTS

a0 = -4.323104e-005
a1 = 2.900539e-004
a2 = -3.074773e-006
a3 = 1.639675e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	579835.0	1.0000	0.0000
4.5000	496028.0	4.5000	-0.0000
15.0000	316707.0	15.0001	0.0001
18.5000	274425.0	18.5000	0.0000
24.0000	220407.0	23.9999	-0.0001
29.0000	181688.0	29.0001	0.0001
32.5000	159239.0	32.4999	-0.0001

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



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Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7325
CALIBRATION DATE: 15-Sep-09

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.034364e+000
h =  1.266892e-001
i = -2.835848e-004
j =  3.533614e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  0.0000e+000
```

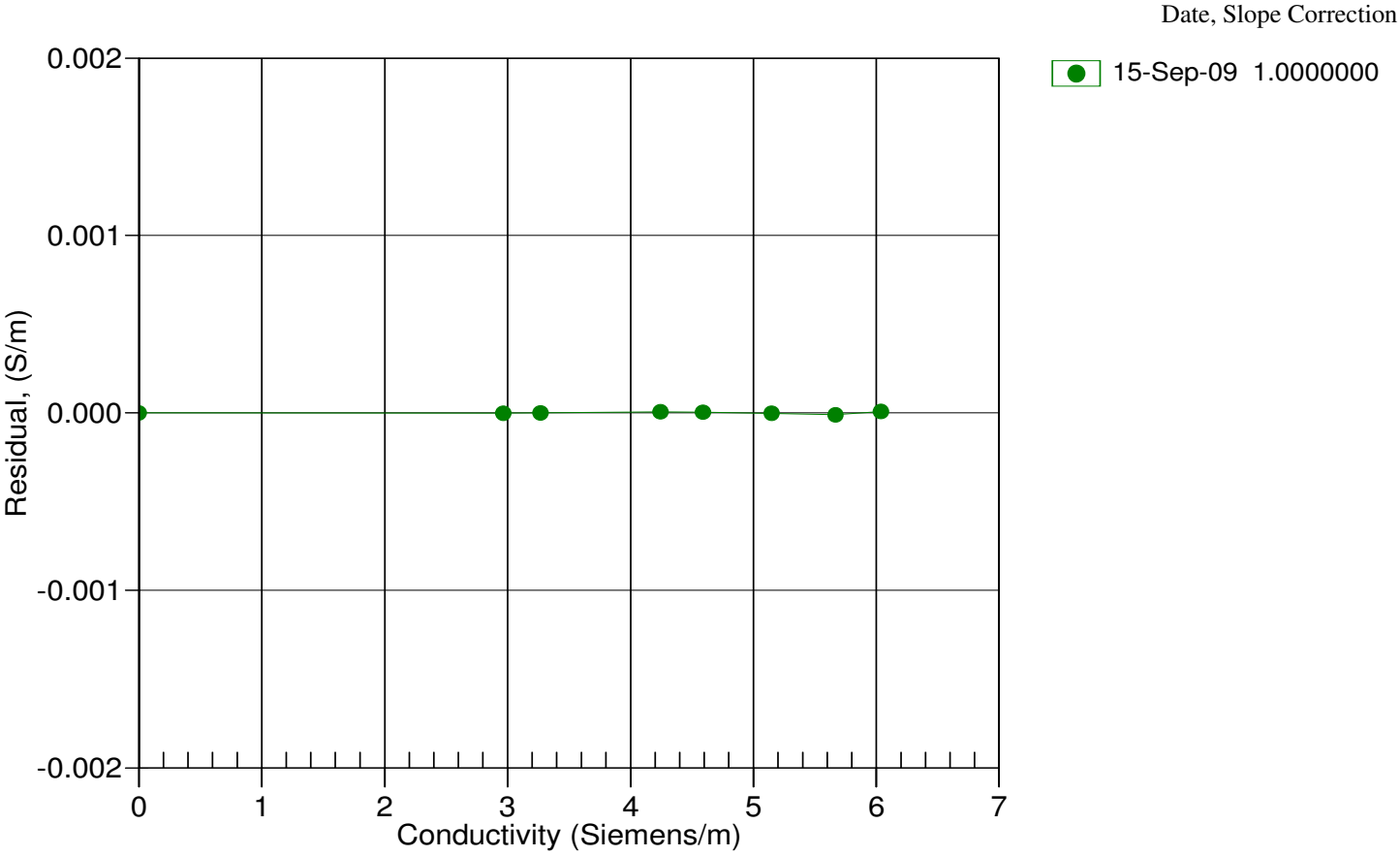
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2863.28	0.00000	0.00000
1.0000	34.6526	2.96330	5627.98	2.96330	-0.00000
4.5000	34.6328	3.26911	5838.71	3.26911	-0.00000
15.0000	34.5895	4.24672	6465.65	4.24673	0.00000
18.5000	34.5800	4.59039	6671.76	4.59040	0.00000
24.0000	34.5691	5.14590	6991.75	5.14590	-0.00000
29.0000	34.5631	5.66550	7277.96	5.66549	-0.00001
32.5000	34.5598	6.03630	7475.31	6.03631	0.00001

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



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Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7325
CALIBRATION DATE: 09-Sep-09

SBE 37 PRESSURE CALIBRATION DATA
1450 psia S/N 2895334

COEFFICIENTS:

PA0 =	-2.522290e-002	PTCA0 =	5.243181e+005
PA1 =	4.730361e-003	PTCA1 =	1.739676e+000
PA2 =	-1.344210e-011	PTCA2 =	-1.237694e-001
PTEMPA0 =	-7.142572e+001	PTCB0 =	2.522750e+001
PTEMPA1 =	4.898920e-002	PTCB1 =	-9.000000e-004
PTEMPA2 =	-2.721073e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FSR
14.62	527395.0	1912.0	14.63	0.00
314.97	590824.0	1935.0	314.87	-0.01
614.88	654217.0	1938.0	614.82	-0.00
914.86	717650.0	1939.0	914.85	-0.00
1214.90	781096.0	1940.0	1214.83	-0.00
1464.86	833994.0	1940.0	1464.86	-0.00
1214.80	781100.0	1939.0	1214.85	0.00
914.78	717652.0	1939.0	914.86	0.01
614.82	654224.0	1939.0	614.86	0.00
314.87	590833.0	1939.0	314.92	0.00
14.63	527395.0	1940.0	14.65	0.00

THERMAL CORRECTION

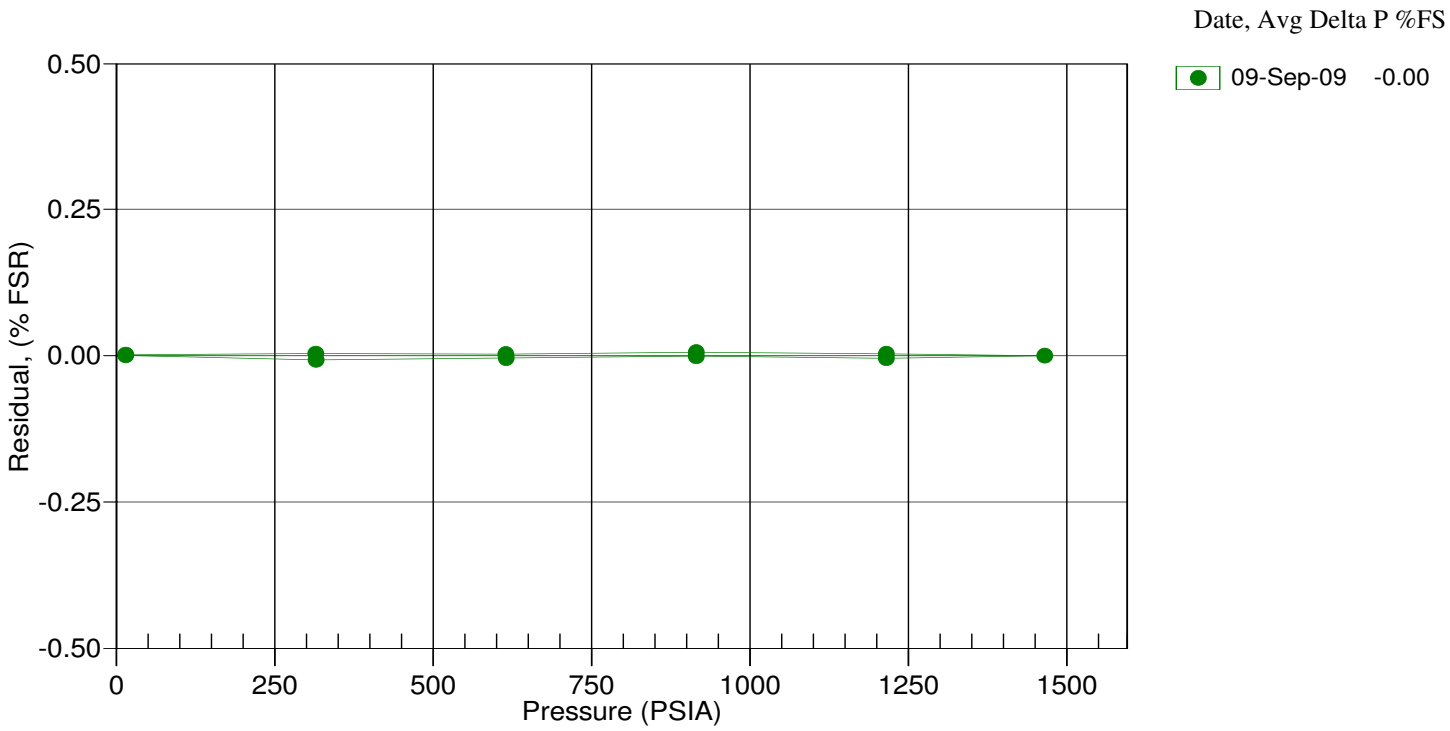
TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2147	527540.00
29.00	2074	527560.01
24.00	1969	527580.82
18.50	1855	527602.60
15.00	1782	527613.15
4.50	1563	527620.75
1.00	1491	527612.87

TEMP(ITS90)	SPAN(mV)
-5.00	25.23
35.00	25.20

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM53517-7326
Pressure Sensor	1000 Dbar Druck, SN 2895335
Firmware Version	V 3.0b / V 2.0a
Memory	8192Kb
Conductivity Range	0-7 S/m
Microcat ID#	05
Zero Conductivity	2634.691 Hz
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**1000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

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Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **8/24/2009** Description **SBE-37 Microcat**

Job Number: **53517** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **7326**

Pressure Sensor Information:

Sensor Type: **Druck**
Sensor Serial Number: **2895335**
Sensor Rating: **1450**

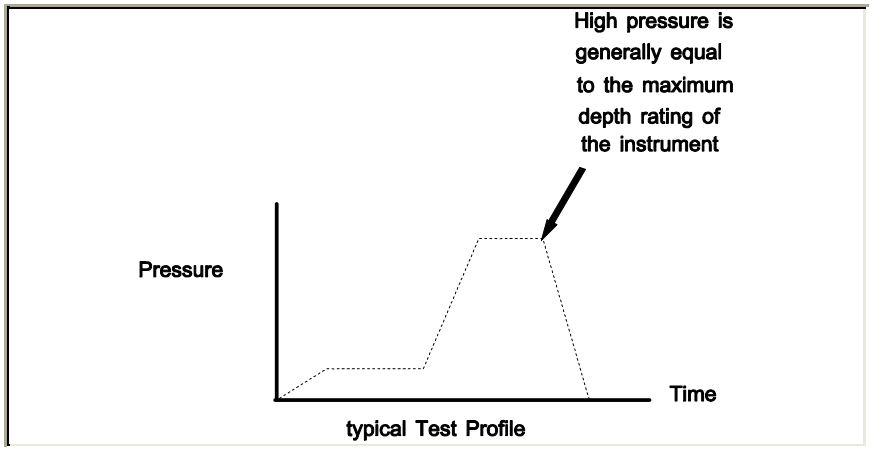
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **1450** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: ND



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7326
CALIBRATION DATE: 15-Sep-09

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

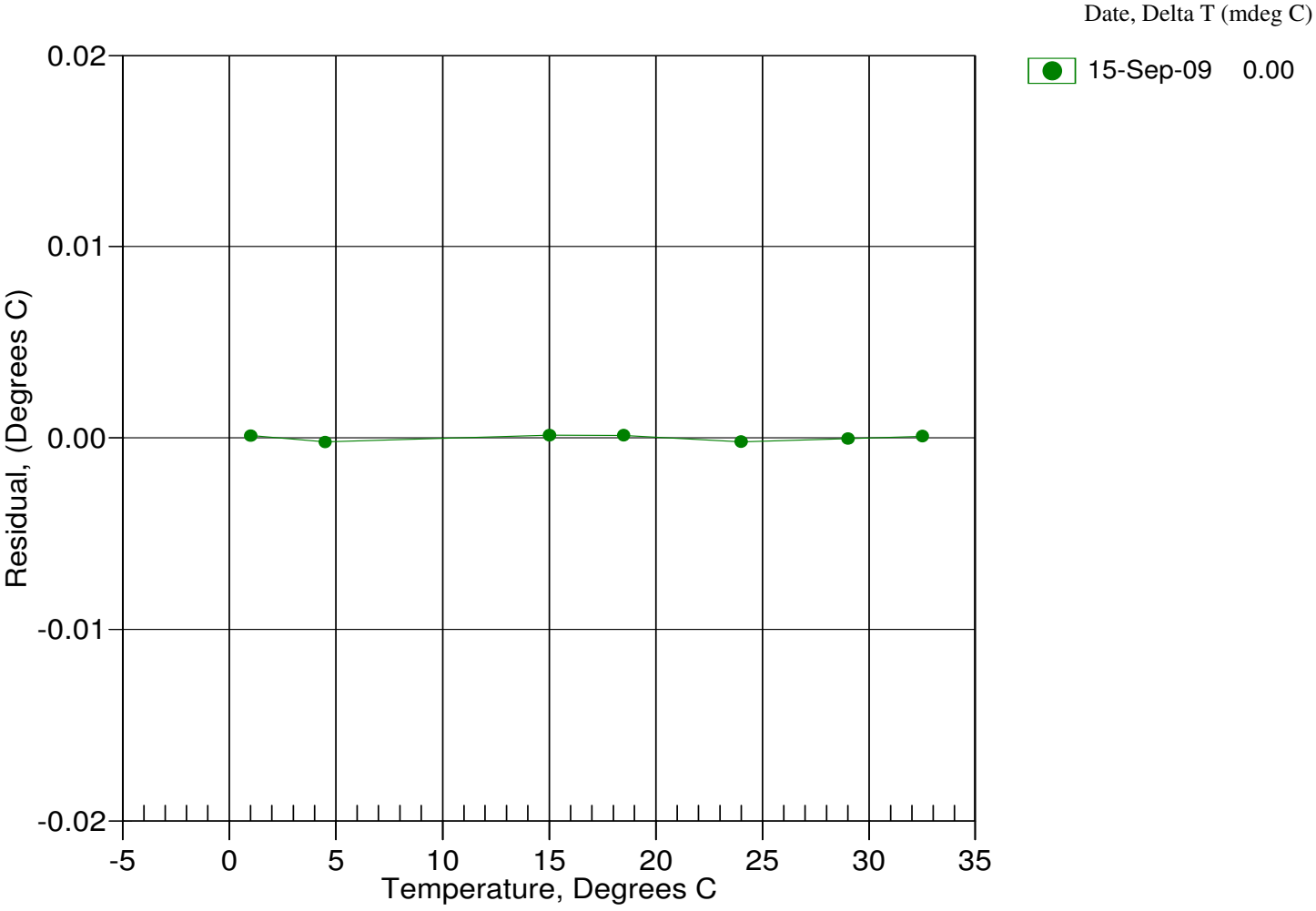
ITS-90 COEFFICIENTS

a0 = -8.534886e-005
a1 = 3.003544e-004
a2 = -3.887301e-006
a3 = 1.842536e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	582018.0	1.0001	0.0001
4.5000	497764.0	4.4998	-0.0002
15.0000	317561.0	15.0001	0.0001
18.5000	275097.0	18.5001	0.0001
24.0000	220868.0	23.9998	-0.0002
29.0000	182013.0	28.9999	-0.0001
32.5000	159490.0	32.5001	0.0001

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 7326
CALIBRATION DATE: 15-Sep-09

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.009723e+000
h =  1.458040e-001
i = -2.222171e-004
j =  3.703589e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  1.2652e-007
```

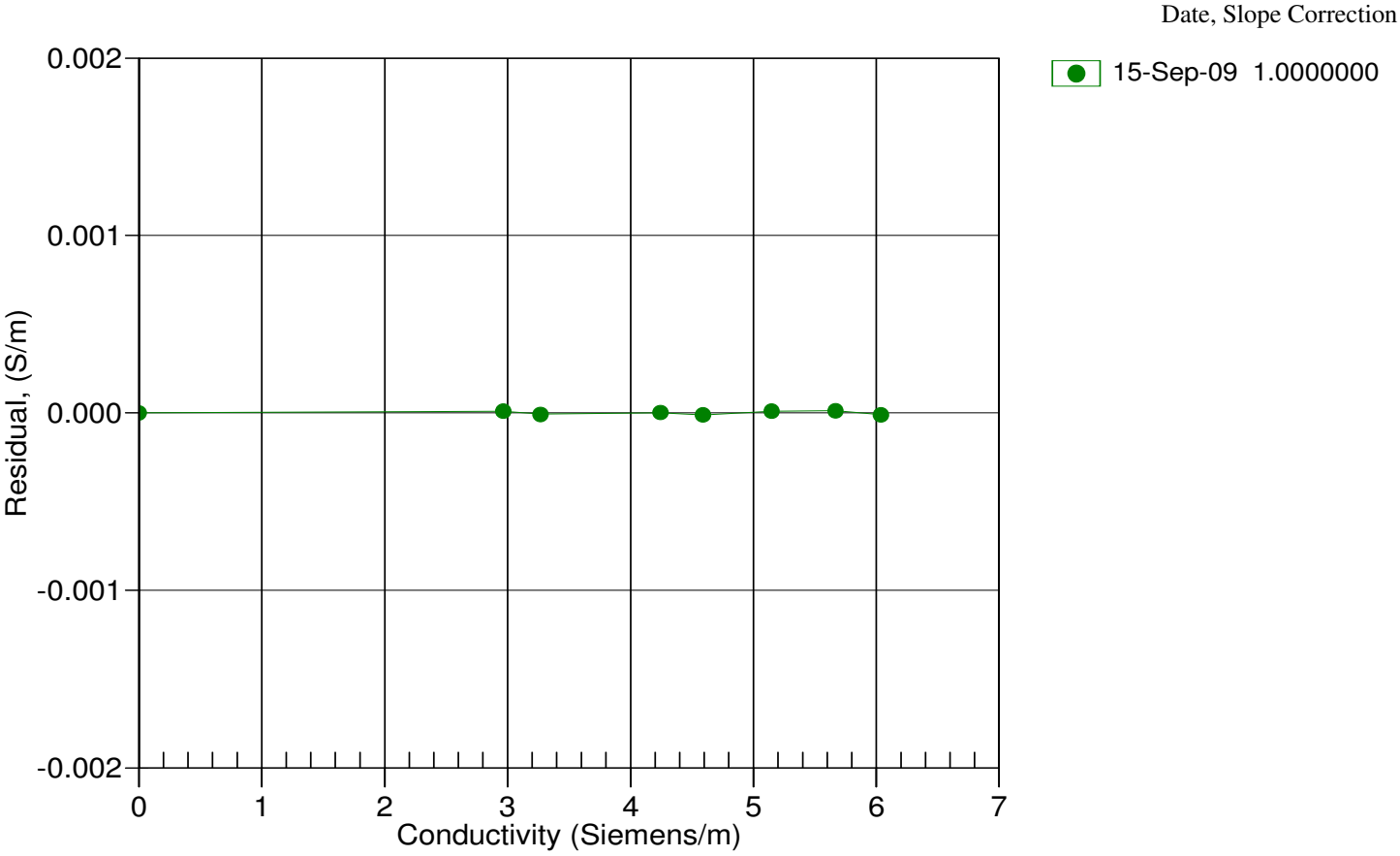
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2634.54	0.00000	0.00000
1.0000	34.6526	2.96330	5222.77	2.96331	0.00001
4.5000	34.6328	3.26911	5419.43	3.26911	-0.00001
15.0000	34.5895	4.24672	6004.38	4.24673	0.00000
18.5000	34.5800	4.59039	6196.64	4.59038	-0.00001
24.0000	34.5691	5.14590	6495.13	5.14591	0.00001
29.0000	34.5631	5.66550	6762.09	5.66551	0.00001
32.5000	34.5598	6.03630	6946.14	6.03629	-0.00001

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7326
CALIBRATION DATE: 09-Sep-09

SBE 37 PRESSURE CALIBRATION DATA
1450 psia S/N 2895335

COEFFICIENTS:

PA0 =	2.627455e-002	PTCA0 =	5.246777e+005
PA1 =	4.918312e-003	PTCA1 =	3.169438e+000
PA2 =	-1.703583e-011	PTCA2 =	-1.249068e-001
PTEMPA0 =	-7.272478e+001	PTCB0 =	2.532988e+001
PTEMPA1 =	4.914815e-002	PTCB1 =	3.750000e-004
PTEMPA2 =	-3.626379e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FSR
14.62	527659.0	1940.0	14.63	0.00
314.97	588740.0	1961.0	314.89	-0.01
614.88	649787.0	1962.0	614.84	-0.00
914.86	710875.0	1963.0	914.87	0.00
1214.90	771980.0	1963.0	1214.85	-0.00
1464.86	822921.0	1963.0	1464.84	-0.00
1214.80	771982.0	1963.0	1214.86	0.00
914.78	710868.0	1965.0	914.83	0.00
614.82	649782.0	1966.0	614.82	-0.00
314.87	588744.0	1966.0	314.91	0.00
14.63	527661.0	1968.0	14.66	0.00

THERMAL CORRECTION

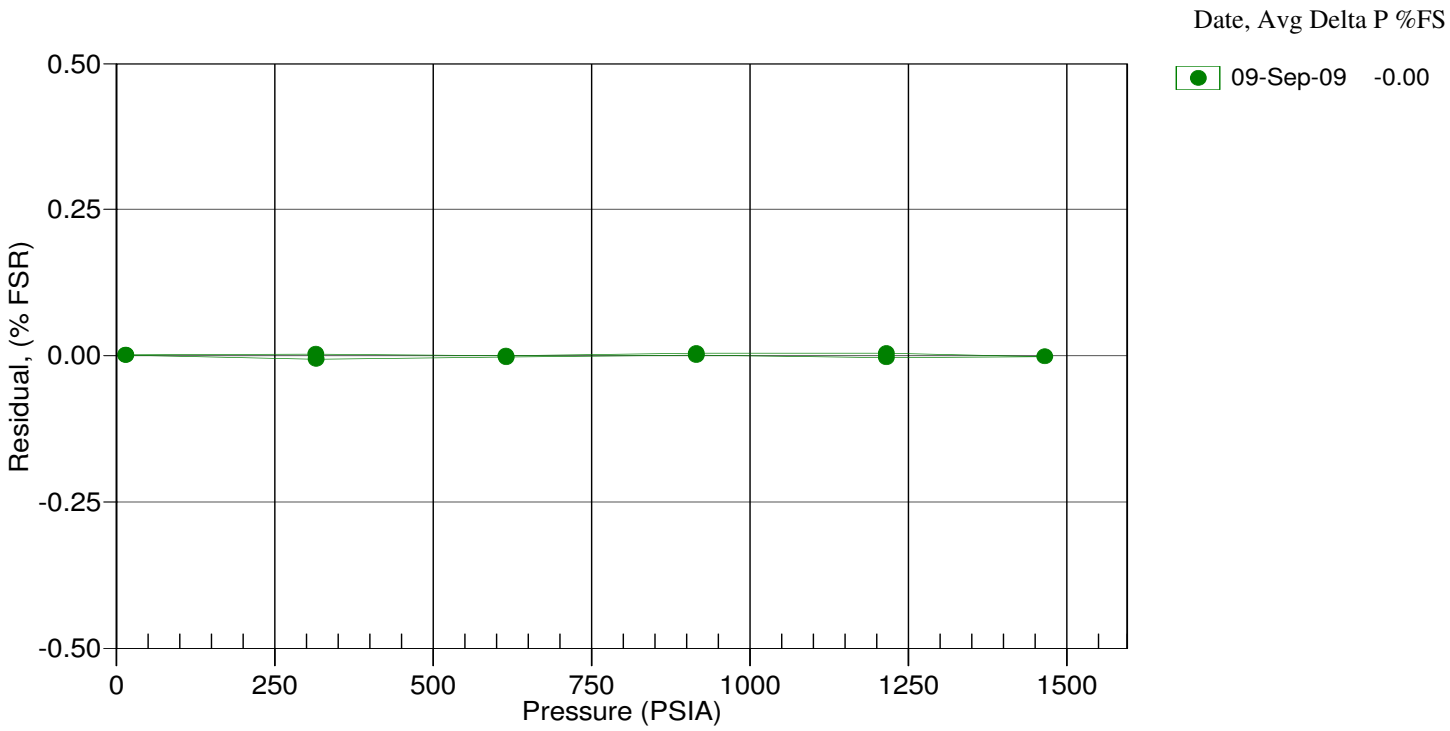
TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2176	527813.00
29.00	2102	527829.05
24.00	1998	527845.07
18.50	1882	527856.93
15.00	1809	527861.63
4.50	1590	527855.25
1.00	1517	527843.44

TEMP(ITS90)	SPAN(mV)
-5.00	25.33
35.00	25.34

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM53517-7327
Pressure Sensor	1000 Dbar Druck, SN 2895336
Firmware Version	V 3.0b / V 2.0a
Memory	8192Kb
Conductivity Range	0-7 S/m
Microcat ID#	06
Zero Conductivity	2565.832 Hz
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**1000 meters**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

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Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **8/24/2009** Description **SBE-37 Microcat**

Job Number: **53517** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **7327**

Pressure Sensor Information:

Sensor Type: **Druck**
Sensor Serial Number: **2895336**
Sensor Rating: **1450**

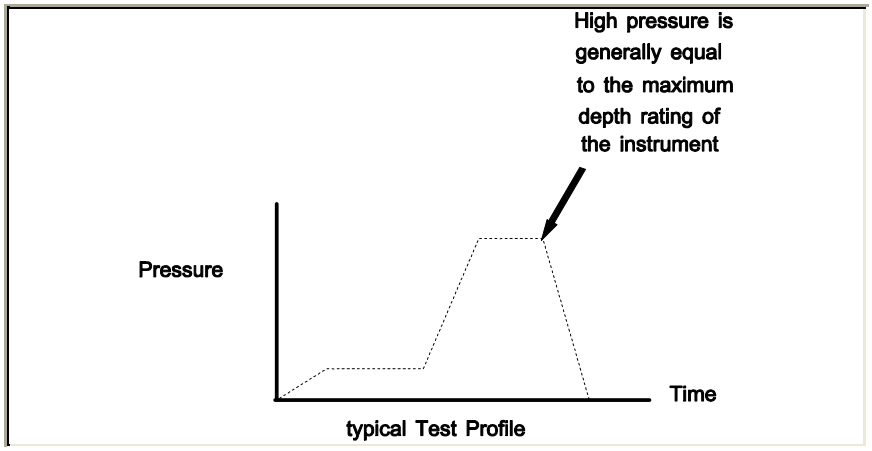
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **1450** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: ND



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7327
CALIBRATION DATE: 15-Sep-09

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

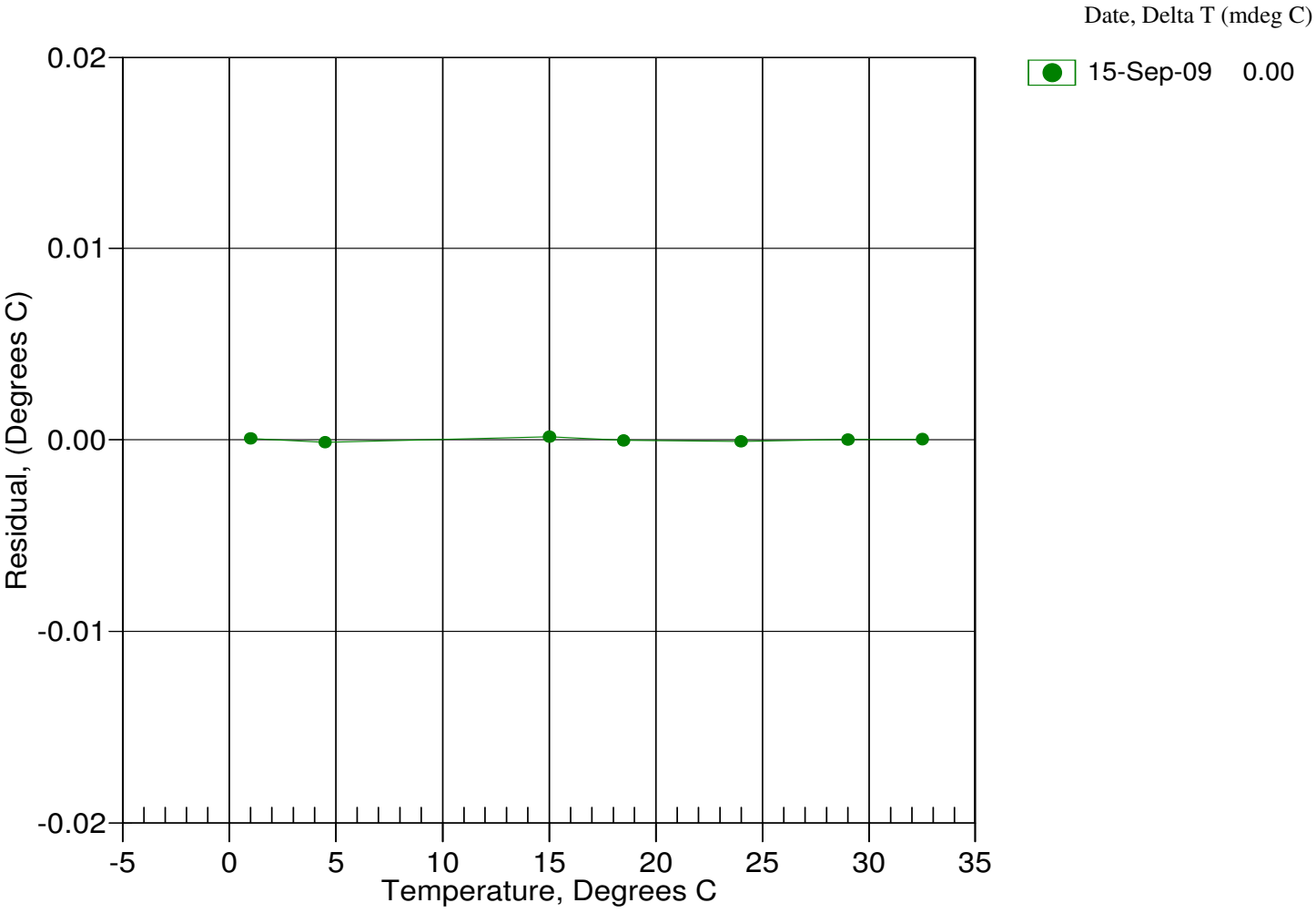
ITS-90 COEFFICIENTS

a0 = -1.259970e-004
a1 = 3.075227e-004
a2 = -4.358725e-006
a3 = 1.980824e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	574639.0	1.0001	0.0001
4.5000	491935.0	4.4999	-0.0001
15.0000	314728.0	15.0002	0.0002
18.5000	272889.0	18.5000	-0.0000
24.0000	219393.0	23.9999	-0.0001
29.0000	181016.0	29.0000	0.0000
32.5000	158748.0	32.5000	0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 7327
CALIBRATION DATE: 15-Sep-09

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.030927e+000
h =  1.567899e-001
i = -1.583513e-004
j =  3.459998e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  1.2991e-007
```

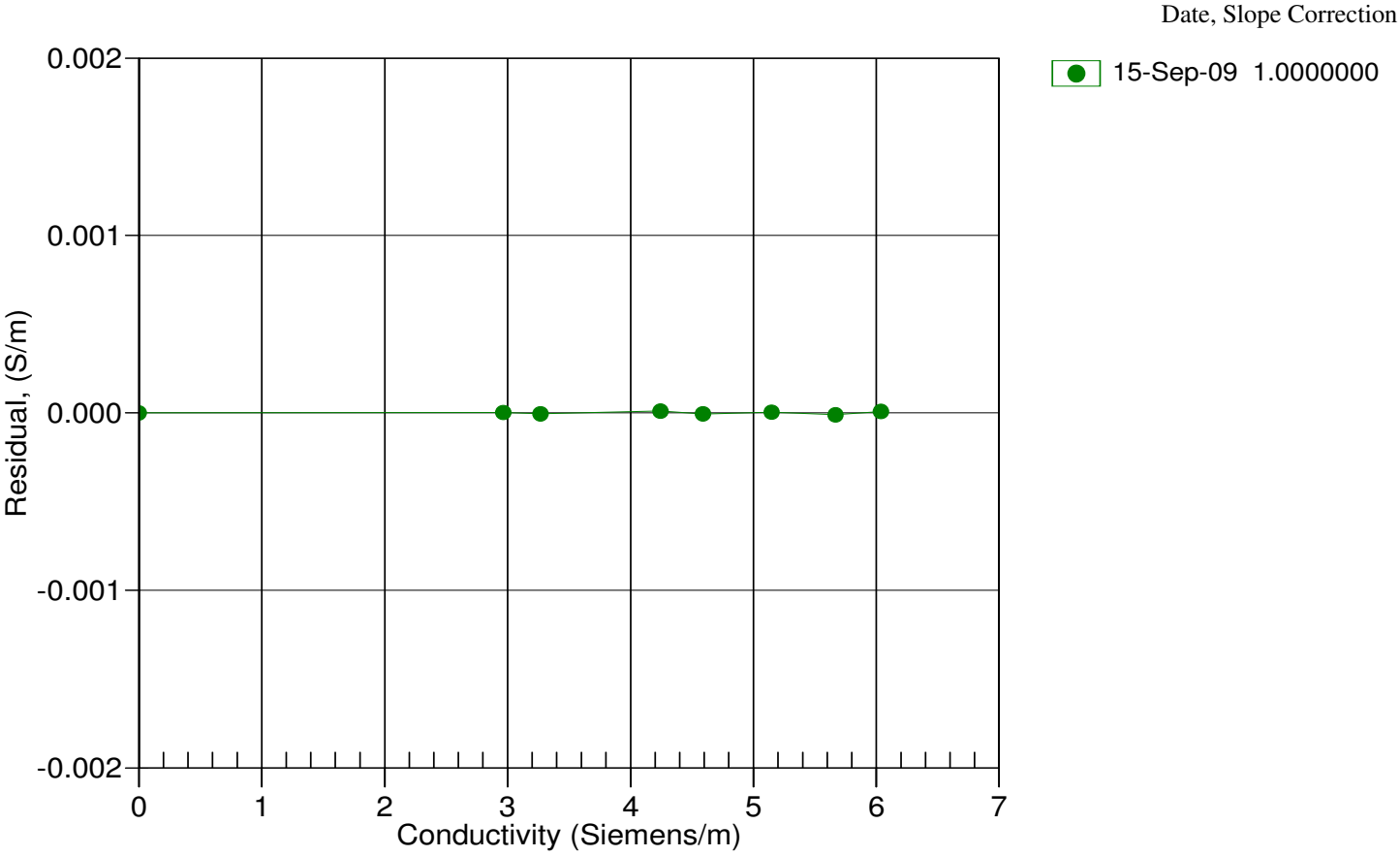
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2565.68	0.00000	0.00000
1.0000	34.6526	2.96330	5045.97	2.96330	0.00000
4.5000	34.6328	3.26911	5234.97	3.26911	-0.00001
15.0000	34.5895	4.24672	5797.37	4.24673	0.00001
18.5000	34.5800	4.59039	5982.29	4.59039	-0.00001
24.0000	34.5691	5.14590	6269.45	5.14591	0.00000
29.0000	34.5631	5.66550	6526.34	5.66548	-0.00001
32.5000	34.5598	6.03630	6703.51	6.03631	0.00001

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 7327
CALIBRATION DATE: 09-Sep-09

SBE 37 PRESSURE CALIBRATION DATA
1450 psia S/N 2895336

COEFFICIENTS:

PA0 =	1.135950e-001	PTCA0 =	5.244640e+005
PA1 =	4.576195e-003	PTCA1 =	3.891227e+000
PA2 =	-1.381943e-011	PTCA2 =	-1.239492e-001
PTEMPA0 =	-7.266061e+001	PTCB0 =	2.526263e+001
PTEMPA1 =	5.026074e-002	PTCB1 =	1.250000e-004
PTEMPA2 =	-6.153766e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	THERMISTOR OUTPUT	COMPUTED PRESSURE	ERROR %FSR
14.62	527665.0	1913.0	14.64	0.00
314.97	593302.0	1937.0	314.92	-0.00
614.88	658892.0	1936.0	614.85	-0.00
914.86	724527.0	1938.0	914.87	0.00
1214.90	790177.0	1938.0	1214.84	-0.00
1464.86	844913.0	1938.0	1464.86	-0.00
1214.80	790175.0	1939.0	1214.83	0.00
914.78	724519.0	1940.0	914.84	0.00
614.82	658887.0	1941.0	614.83	0.00
14.63	527665.0	1943.0	14.65	0.00

THERMAL CORRECTION

TEMP ITS90	THERMISTOR OUTPUT	INST OUTPUT
32.50	2149	527836.00
29.00	2075	527847.90
24.00	1971	527860.55
18.50	1856	527868.25
15.00	1783	527870.85
4.50	1565	527857.23
1.00	1493	527841.28

TEMP(ITS90)	SPAN(mV)
-5.00	25.26
35.00	25.27

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

pressure (psia) = PA0 + PA1 * n + PA2 * n^2

